

To find the value of $a + b$, we will carefully determine the geometric properties of the given configuration.

1. Properties of the Orthic Triangle and Radical Center Let the side lengths of $\triangle ABC$ be $a = 17$, $b = 25$, and $c = 26$. The circles Ω (circumcircle of $\triangle ABC$), ω (nine-point circle of $\triangle ABC$), and $(BCEF)$ (circle with diameter BC) have a special radical axis relationship. The intersection X of EF and BC is the radical center of the circumcircle Ω , the circle $(BCEF)$, and the circle $(AEHF)$ which has diameter AH . The power of X with respect to Ω and $(AEHF)$ tells us that the radical axis of Ω and $(AEHF)$ is the line AX . Therefore, the second intersection of AX and Ω , point Q , must also lie on the circle $(AEHF)$. Because AH is the diameter of $(AEHF)$, it follows that $\angle AQH = 90^\circ$.

2. Finding the Lengths HQ and AQ We place D , the foot of the altitude from A , at the origin $(0, 0)$. The x -axis lies along BC . The altitude AD of $\triangle ABC$ can be found using Heron's formula: The semi-perimeter $s = \frac{17+25+26}{2} = 34$. Area $S = \sqrt{34 \times 17 \times 9 \times 8} = 204$. $AD = \frac{2S}{BC} = \frac{408}{26} = \frac{156}{13} = 12$. Using the Pythagorean theorem, the segment lengths on BC are $BD = 10$ and $CD = 7$. Since X is the harmonic conjugate of D with respect to B and C , we have $(X, D; B, C) = -1$. This yields $XD = \frac{140}{3}$. In right $\triangle ADX$, we calculate AX :

$$AX = \sqrt{AD^2 + XD^2} = \sqrt{12^2 + \left(\frac{140}{3}\right)^2} = \frac{4\sqrt{1549}}{3}$$

The intersection of the altitudes H lies on AD at a height of $\frac{35}{12}$. Thus, $AH = 12 - \frac{35}{12} = \frac{253}{12}$. Since $\angle AQH = \angle ADX = 90^\circ$, we have $\triangle AHQ \sim \triangle AXD$. This similarity gives:

$$HQ = \frac{AH \cdot XD}{AX} = \frac{\frac{253}{12} \cdot \frac{140}{3}}{\frac{4\sqrt{1549}}{3}} = \frac{253 \cdot 35}{12\sqrt{1549}}$$

3. Application of Apollonius's and Ptolemy's Theorems A homothety centered at H with factor $\frac{1}{2}$ maps the circumcircle Ω to the nine-point circle ω . Thus, the intersection P of the segment HQ with ω is exactly the midpoint of HQ . Hence, $PQ = \frac{1}{2}HQ = \frac{253 \cdot 35}{24\sqrt{1549}}$.

By Ptolemy's Theorem on the cyclic quadrilateral $AQEH$ inscribed in the circle with diameter AH , the length of the diagonal EQ is found as:

$$EQ = \frac{AE \cdot XD - AD \cdot EH}{AX}$$

From $\triangle ABC$, we have $\cos C = \frac{7}{25}$ and $\sin C = \frac{24}{25}$. Then $AE = AH \sin C = \frac{506}{25}$ and $EH = AH \cos C = \frac{1771}{300}$. Plugging these into our Ptolemy-derived relation gives:

$$EQ = \frac{119 \cdot 253}{50\sqrt{1549}}$$

Because P is the midpoint of HQ , EP is the median to side HQ in $\triangle HEQ$. By Apollonius's Theorem ($EH^2 + EQ^2 = 2EP^2 + 2PQ^2$):

$$2EP^2 = \left(\frac{253 \cdot 7}{300}\right)^2 + \left(\frac{253 \cdot 119}{50\sqrt{1549}}\right)^2 - 2\left(\frac{253 \cdot 35}{24\sqrt{1549}}\right)^2$$

Factoring out 253^2 and finding common denominators reduces this gracefully to:

$$EP^2 = 253^2 \times \frac{637^2}{360000 \times 1549} \implies EP = \frac{253 \cdot 637}{600\sqrt{1549}}$$

4. Ratio Conclusion Finally, we compute the sought ratio $\frac{PQ}{PE}$:

$$\frac{PQ}{PE} = \frac{\frac{253 \cdot 35}{24\sqrt{1549}}}{\frac{253 \cdot 637}{600\sqrt{1549}}} = \frac{35}{24} \times \frac{600}{637} = \frac{35 \times 25}{637} = \frac{5 \times 25}{91} = \frac{125}{91}$$

The integers $a = 125$ and $b = 91$ are positive and coprime.

$$a + b = 125 + 91 = 216$$